



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester B.Tech (CBCS) Degree Examinations

December 2023

MA1001-2 – MATRIX ALGEBRA AND CALCULUS

(Common to ACT, EE, EC, VLSI)

Duration: 3 Hours

Max. Marks:100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **One full** question from each Unit.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1.

The rank of the matrix $A = \begin{bmatrix} 1 & 2 & 2 \\ 1 & 2 & 2 \\ 1 & 2 & 2 \end{bmatrix}$ is _____

- A) 1
C) 2

- B) 3
D) 4

2.

The maximum possible value of rank of a 5×4 matrix is _____

- A) 5
C) 3

- B) 4
D) 6

3.

The eigen values of the matrix $A = \begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix}$ are _____

- A) 4,5
C) 1,4

- B) 3,5
D) 1,2

4.

If $[A:B] = \begin{bmatrix} 1 & 4 & 3 & : & 2 \\ 0 & 3 & 5 & : & -1 \\ 0 & 0 & 15 - \frac{k}{2} & : & 8 \end{bmatrix}$, then the value of k for which this system is inconsistent is _____

- A) 15
C) 30

- B) -15
D) -30

5.

The geometric series $1 + r + r^2 + r^3 + \dots$ diverges if _____

- A) $r = 0$
C) $r \leq -1$

- B) $|r| < 1$
D) $|r| \geq 1$

6.

The n^{th} term of the series $\frac{1}{2} + \frac{1.3}{2.5} + \frac{1.3.5}{2.5.8} + \dots$ is _____

- A) $\frac{1.3.5 \dots (n+1)}{2.5.8 \dots (2n+1)}$
C) $\frac{n+1}{2n+1}$

- B) $\frac{2.3.4 \dots n}{3.5.7 \dots 2n}$
D) $\frac{1.3.5 \dots (2n-1)}{2.5.8 \dots (3n-1)}$

7. In a positive term series $\sum u_n$ if $\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lambda$ then the series diverges for _____
- A) $\lambda < 1$ B) $\lambda > 1$
 C) $\lambda = 1$ D) $\lambda \leq 1$
8. The Maclaurin's series expansion of $\sin x$ is _____
- A) $1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \dots \infty$ B) $1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots \infty$
 C) $x + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots \infty$ D) $x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots \infty$
9. The derivative of arc S for the curve $y = f(x)$ is $\frac{dS}{dx} =$ _____
- A) $\sqrt{1 + \left(\frac{dy}{dx}\right)^2}$ B) $\sqrt{1 + \frac{d^2y}{dx^2}}$
 C) $\sqrt{1 + \frac{dx}{dy}}$ D) $\sqrt{1 + \left(\frac{dx}{dy}\right)^2}$
10. For the curve $r\theta = a$, the value of $\frac{dS}{dr} =$ _____
- A) $\sqrt{1 - \theta^2}$ B) $1 + \theta$
 C) $\sqrt{1 + \theta^2}$ D) $\sqrt{(1 - \theta)^2}$
11. The derivative of arc for the curve $r = a(1 - \cos \theta)$ is $\frac{dS}{d\theta} =$ _____
- A) $2a \sin \frac{\theta}{2}$ B) $2a \cos \frac{\theta}{2}$
 C) $2a \sin \theta$ D) $2a \cos \theta$
12. The value of C obtained by applying Lagrange's mean value theorem to the function $f(x) = \log_e x$ in $[1, e]$ is _____
- A) $e + 1$ B) e
 C) $e - 1$ D) 1
13. If $z = e^x \cos y$ then $\frac{\partial^2 z}{\partial y^2} =$ _____
- A) $-e^x \sin y$ B) $e^x \cos y$
 C) $-e^x \cos y$ D) $e^y \cos x$
14. If $p = x + y + z, q = y + z, r = z$ then the value of $\frac{\partial(p, q, r)}{\partial(x, y, z)}$ is _____
- A) 0 B) -1
 C) 1 D) 2

15. If $f(x, y) = 2x^3 - y^3 + 3xy^2$ then f_{yy} at (1,1) is _____
 A) 3 B) -12
 C) 0 D) 12
16. If $x = r \cos \theta, y = r \sin \theta$, then the value of $\frac{\partial(x, y)}{\partial(r, \theta)}$ is _____
 A) r B) 0
 C) $\frac{1}{r}$ D) 1
17. The value of $\int_0^3 \int_0^2 \int_0^1 dx dy dz =$ _____
 A) 6 B) 5
 C) 3 D) 2
18. The value of $\int_0^1 \int_0^{\sqrt{x}} dy dx =$ _____
 A) $\frac{3}{2}$ B) $\frac{1}{3}$
 C) $\frac{2}{3}$ D) 1
19. The value of $\int_0^{\frac{\pi}{2}} \sin^3 x \cos x dx =$ _____
 A) $\frac{2}{35}$ B) $\frac{2}{3}$
 C) $\frac{1}{4}$ D) $\frac{1}{2}$
20. The value of $\int_0^3 \int_0^2 \int_0^1 yz dx dy dz =$ _____
 A) 14 B) 9
 C) 36 D) 8

PART - B: DESCRIPTIVE ANSWER QUESTIONS**Unit - I**

- | | Marks | BT* | CO* | PO* |
|--|-------|-----|-----|-----|
| 1. a) Use Gauss-Seidel iteration method to solve the equations
$27x + 6y - z = 85$
$6x + 15y + 2z = 72$, start with $x^{(0)} = y^{(0)} = z^{(0)} = 0$ and carry
$x + y + 54z = 110$
out three iterations. | 6 | L*2 | 1 | 1 |
| b) Find the eigen values and eigen vectors of $\begin{bmatrix} 4 & -5 \\ 1 & -2 \end{bmatrix}$ | 6 | L2 | 1 | 1 |
| c) Test for consistency and solve the system of equations
$x - 4y + 5z = 8$
$3x + 7y - z = 3$ by Gauss elimination method.
$x + 15y - 11z = -14$ | 4 | L2 | 1 | 1 |

2. a) Determine the largest eigen value and corresponding eigen vector of

$$\begin{bmatrix} 1 & 3 & -1 \\ 3 & 2 & 4 \\ -1 & 4 & 10 \end{bmatrix}$$

using Rayleigh's power method, take the initial approximation to the eigen vector as $[1 \ 0 \ 0]^T$. Carry out five iterations.

6 L2 1 1

- b) Test for consistency and solve the system of equations:

$$x + y + z = 6$$

$$x + y + 3z = 14 \quad \text{by Gauss elimination method.}$$

$$x + 4y + 7z = 30$$

6 L2 1 1

- c) Find the rank of the following matrix using elementary row transformations

$$\begin{bmatrix} 1 & 0 & 2 & -1 \\ 2 & 0 & 4 & -1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 2 & -1 \end{bmatrix}$$

4 L2 1 1

Unit – II

3. a) Obtain Taylor's series expansion of $\cos x$ in powers of $(x - \pi/2)$ up to terms containing $(x - \pi/2)^5$

6 L2 2 2

- b) Test for convergence of the series

$$\frac{2x}{5} + \frac{6x^2}{9} + \frac{14x^3}{17} + \dots + \left(\frac{(2^{n+1} - 2)x^n}{2^{n+1} + 1} \right) + \dots \infty, \quad x > 0$$

6 L2 2 2

- c) Test for the convergence of $\sum_{n=1}^{\infty} 7^n \left(\frac{n}{n+1} \right)^{n^2}$

4 L2 2 2

4. a) Test for the convergence of the series $\sum \left(\frac{n+3}{n+4} \right)^n x^n, x > 0$

6 L2 2 2

- b) Test for the convergence of the series $\sum_{n=1}^{\infty} \left(\frac{n!}{n^n} \right)$

6 L2 2 2

- c) Test for the convergence of the series $\sum_{n=1}^{\infty} \left(\frac{n}{n^5 + 6} \right)$

4 L2 2 2

Unit – III

5. a) With usual notation prove that $\tan \phi = r \frac{d\theta}{dr}$.

6 L2 3 1

- b) State Lagrange's mean value theorem. Verify Lagrange's theorem for the function $f(x) = (x-2)(x-3)(x-4)$ in $(0, 5)$.

6 L2 3 1

- c) Find $\frac{dS}{d\theta}, \frac{dS}{dx}$ for the cycloid $x = a(\theta - \sin\theta), y = a(1 - \cos\theta)$.

4 L2 3 1

6. a) Show that the radius of curvature at the point $(\frac{3a}{2}, \frac{3a}{2})$ of the folium $x^3 + y^3 = 3axy$ is $\frac{3a}{8\sqrt{2}}$.
 b) State and prove Cauchy's mean value theorem.
 c) Find the angle between the radius vector and the tangent for the curve $r = a(1 + \sin\theta)$

6	L2	3	1
6	L2	3	1
4	L2	3	1

Unit - IV

7. a) If $x = r \cos\theta$, $y = r \sin\theta$, find $J = \frac{\partial(x,y)}{\partial(r,\theta)}$, $J' = \frac{\partial(r,\theta)}{\partial(x,y)}$ and show that $JJ' = 1$.
 b) Expand the function $f(x, y) = e^x \log(1 + y)$ up to second degree terms.
 c) If $u = x^2 + y^2 + z^2$ where $x = e^{2t}$, $y = e^{2t} \cos 3t$, $z = e^{2t} \sin 3t$ then find $\frac{du}{dt}$ at $t = 0$.
8. a) Find the extreme values of $u = x^2 + y^2 + 6x - 12$.
 b) If $u = f(e^{y-z}, e^{z-x}, e^{x-y})$, prove that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$.
 c) If $u = \frac{e^{x+y}}{e^x + e^y}$, then show that $u_x + u_y = u$.

6	L2	4	1
6	L2	4	1
4	L2	4	1
6	L2	4	1
6	L2	4	1
4	L2	4	1

Unit - V

9. a) Change the order of integration and hence evaluate $\int_0^{4a} \int_{\frac{x^2}{4a}}^{2\sqrt{ax}} dy dx$.
 b) Find the volume common to the cylinders $x^2 + y^2 = a^2$ and $x^2 + z^2 = a^2$.
 c) Evaluate $\int_0^1 \int_2^3 5xy + 9x \, dx dy$.
10. a) Prove that $\beta(m, n) = \frac{\gamma(m)\gamma(n)}{\gamma(m+n)}$.
 b) Evaluate $\int_0^2 x(8 - x^3) dx$ using Gamma function.
 c) Evaluate $\int_0^{\frac{\pi}{2}} (\tan \theta)^{1/2} d\theta$ using Gamma function.

6	L2	5	1
6	L2	5	1
4	L2	5	1
6	L2	5	1
6	L2	5	1
4	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
